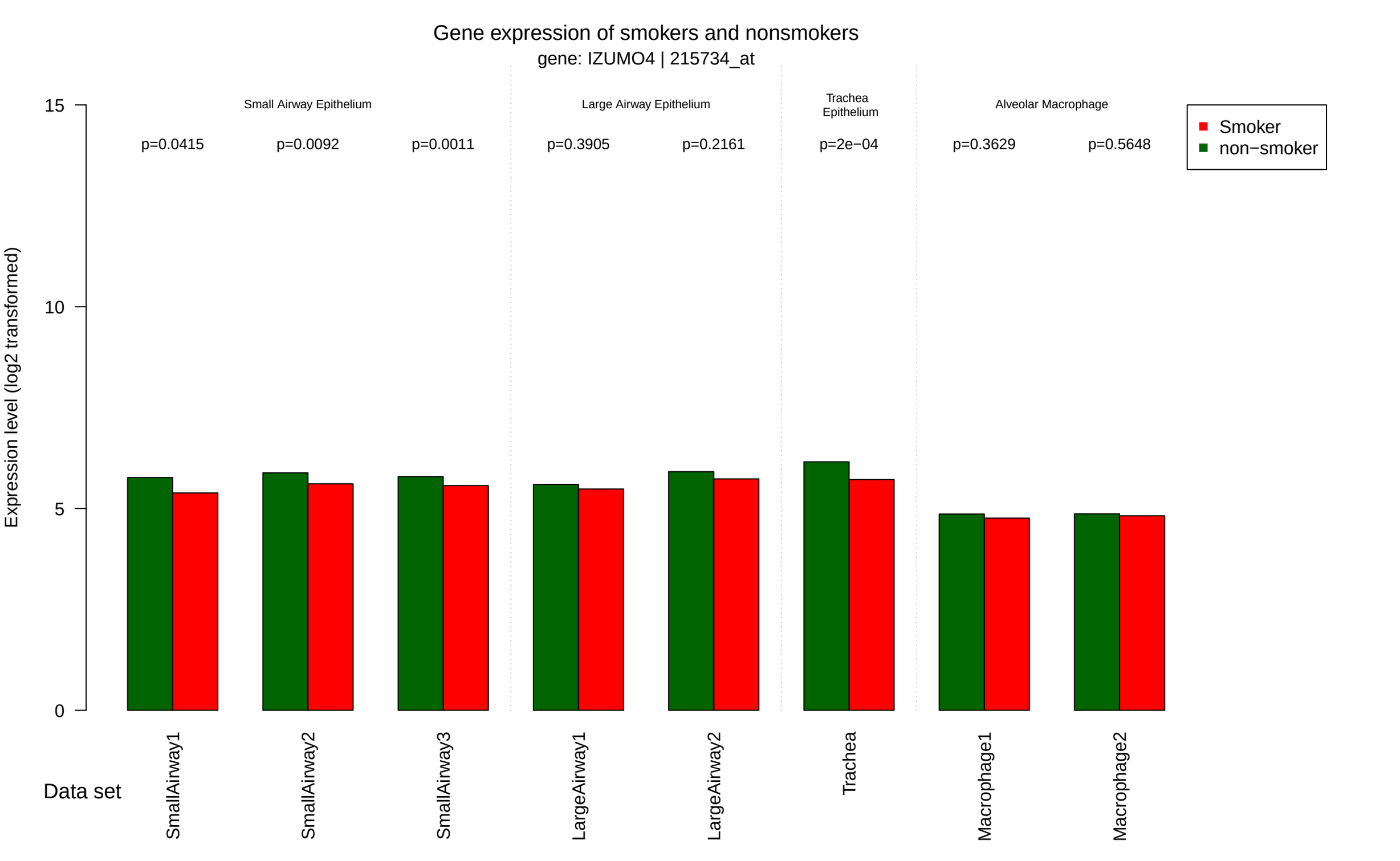
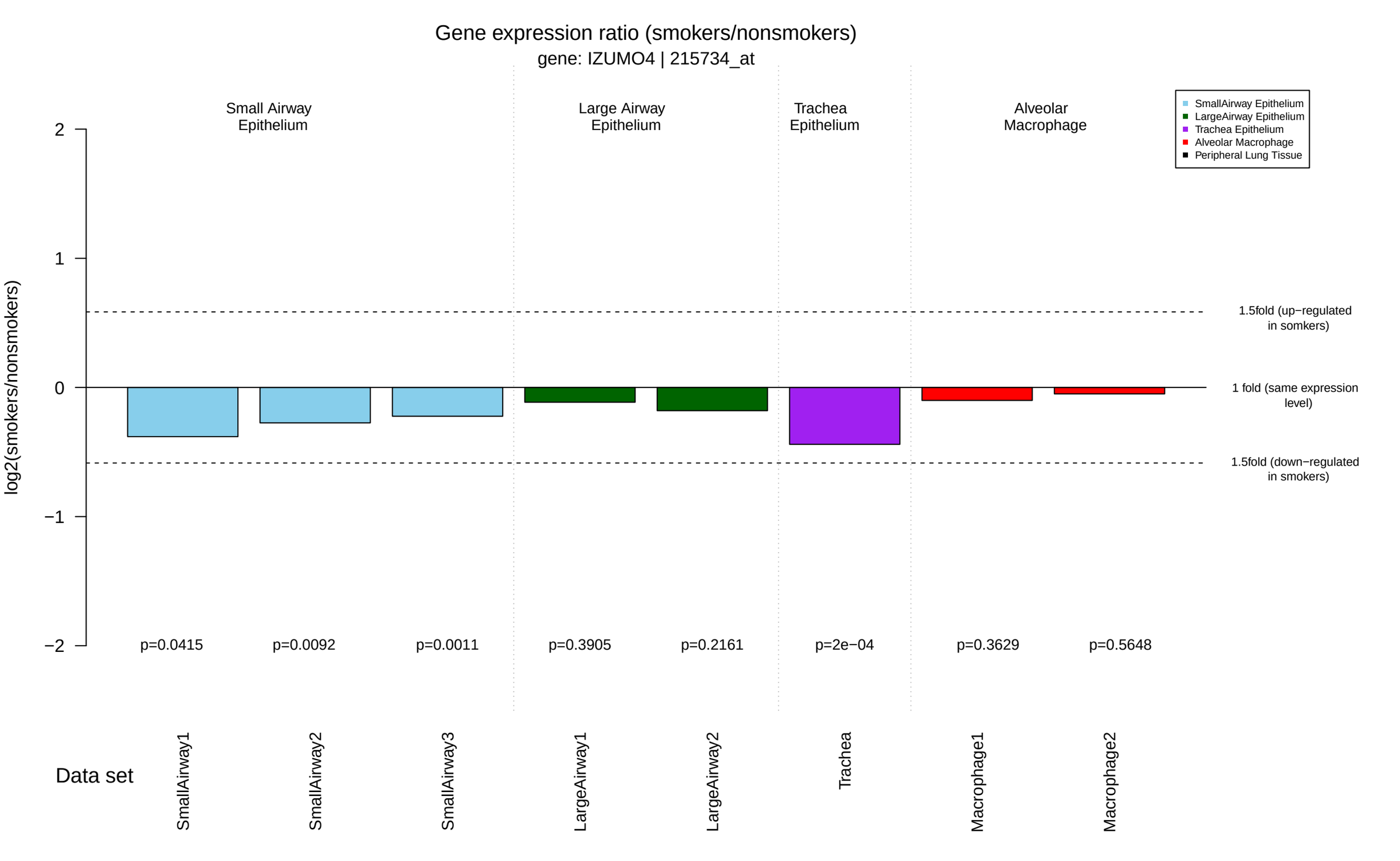
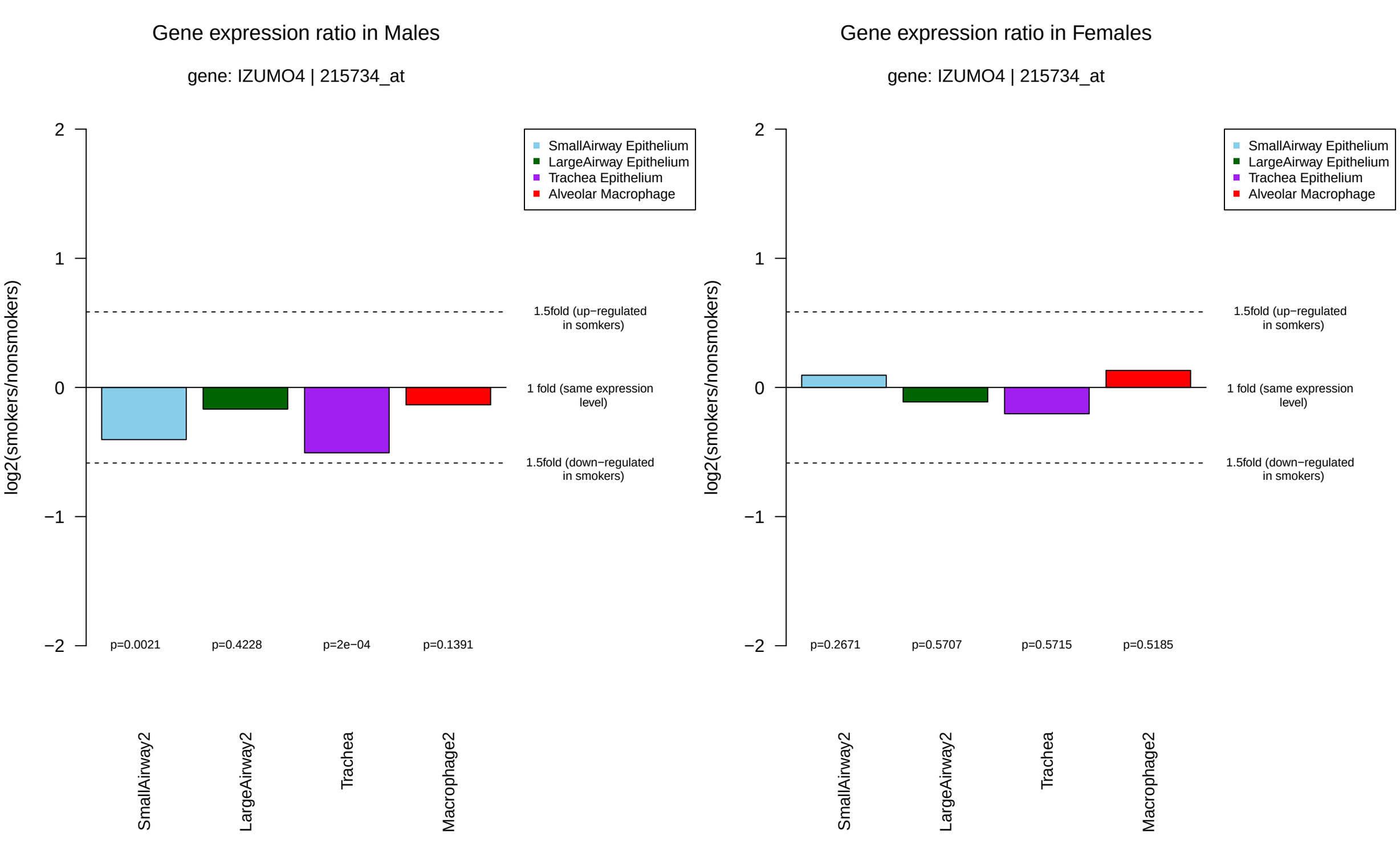
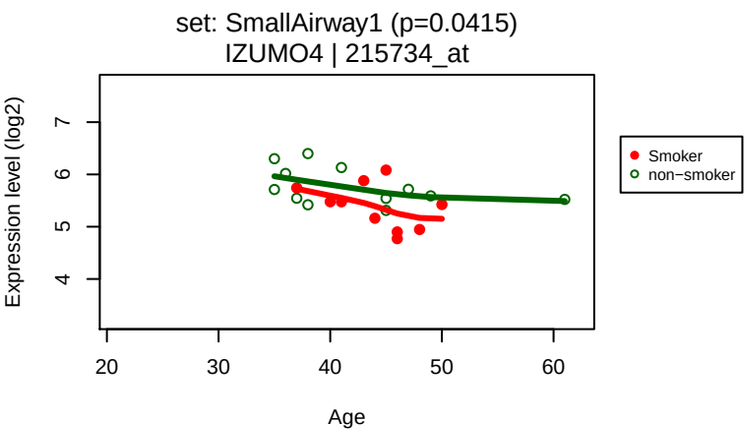
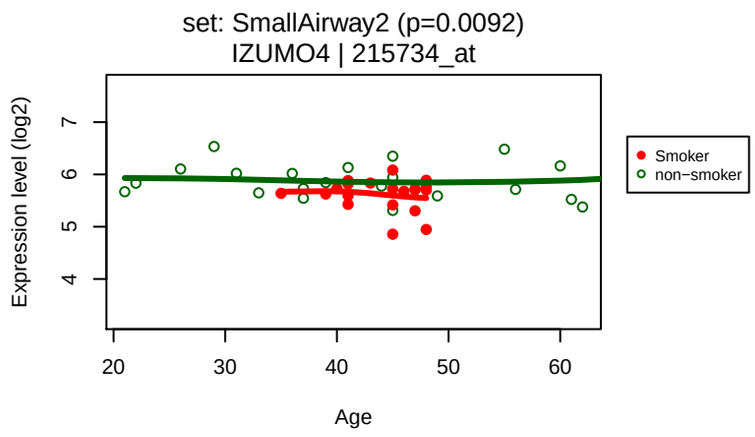
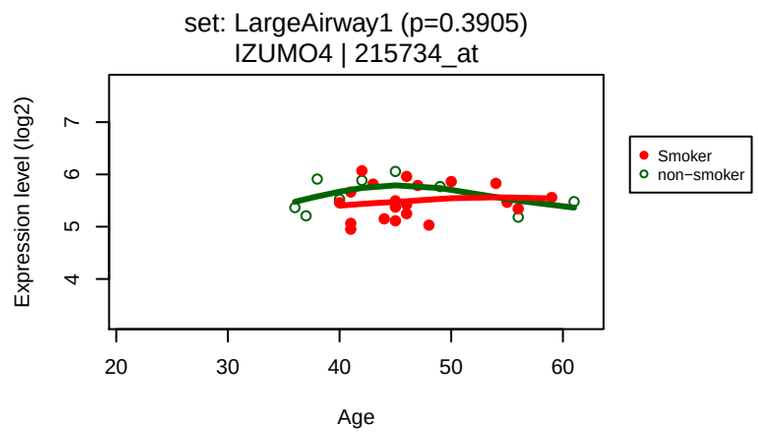

Gene expression of smokers and nonsmokers
gene: IZUMO4 | 215734_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0415
p=0.0092
p=0.0011
p=0.3905
p=0.2161
p=2e−04
p=0.3629
p=0.5648
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

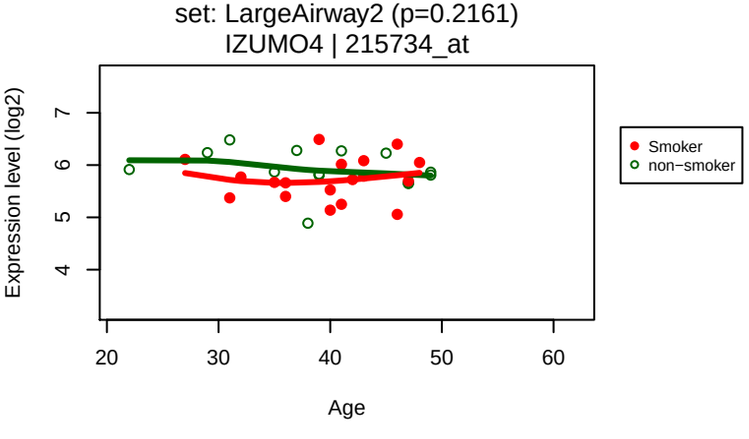

Gene expression ratio (smokers/nonsmokers)
gene: IZUMO4 | 215734_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0415
p=0.0092
p=0.0011
p=0.3905
p=0.2161
p=2e−04
p=0.3629
p=0.5648
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

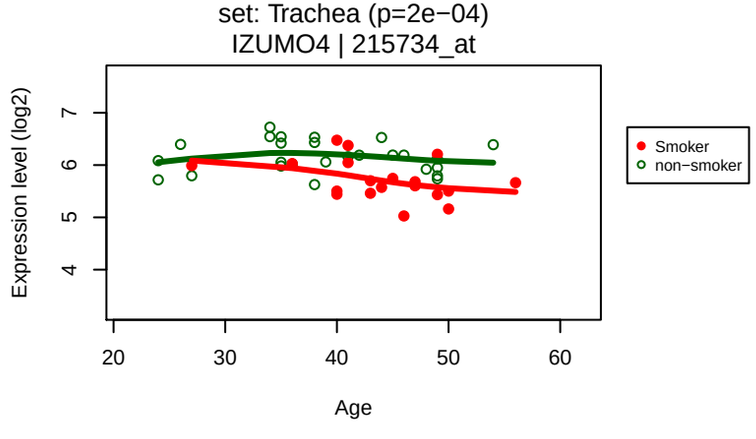

Gene expression ratio in Males
gene: IZUMO4 | 215734_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0021
p=0.4228
p=2e−04
p=0.1391
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IZUMO4 | 215734_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2671
p=0.5707
p=0.5715
p=0.5185
SmallAirway2
LargeAirway2
Trachea
Macrophage2

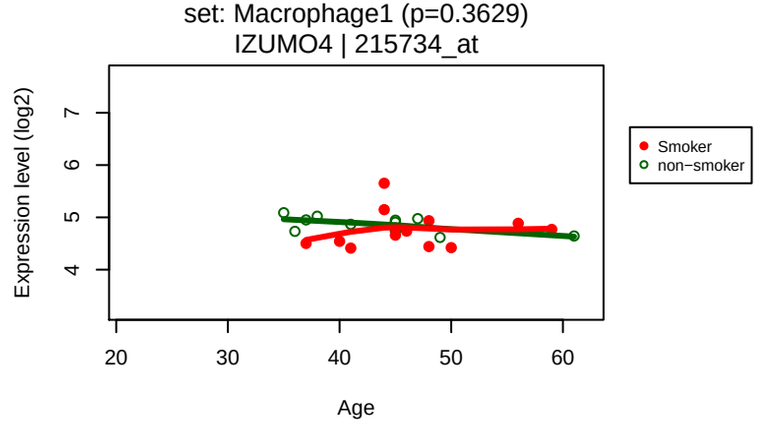

set: SmallAirway1 (p=0.0415)
IZUMO4 | 215734_at
Expression level (log2)
Age
Smoker
non−smoker

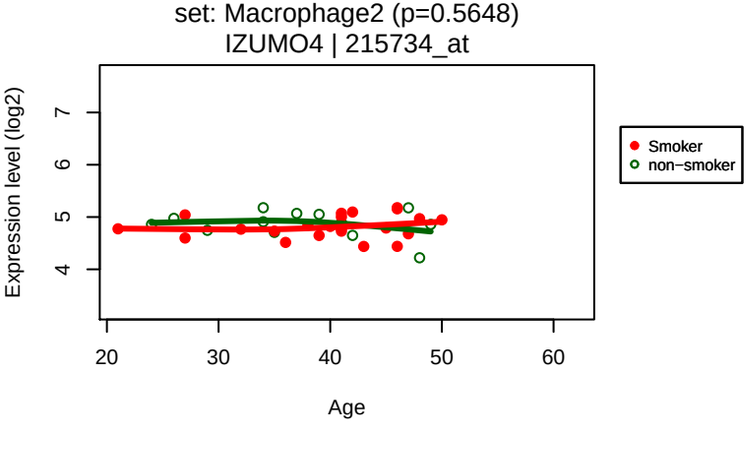

set: SmallAirway2 (p=0.0092)
IZUMO4 | 215734_at
Expression level (log2)
Age
Smoker
non−smoker

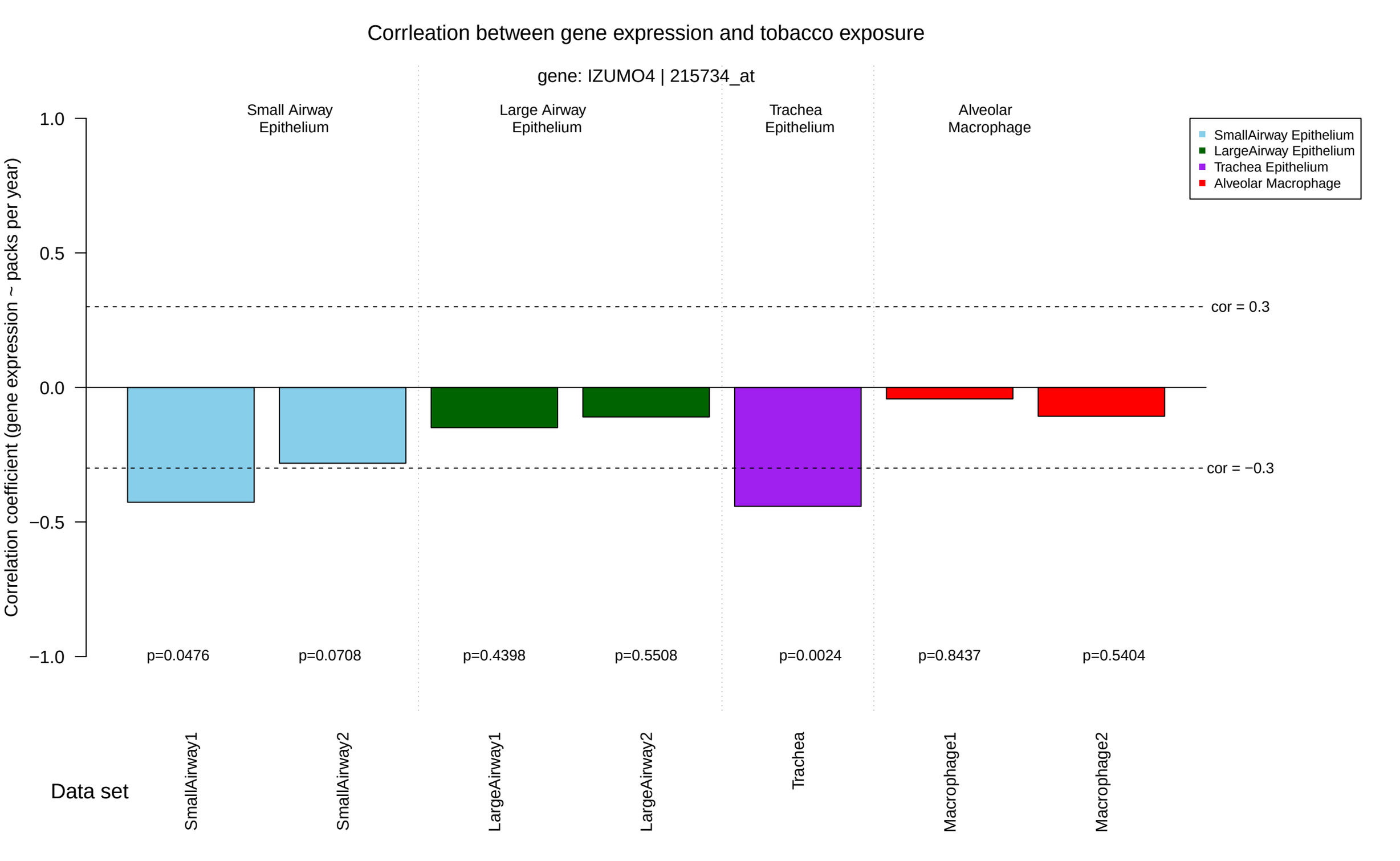

set: LargeAirway1 (p=0.3905)
IZUMO4 | 215734_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2161)
IZUMO4 | 215734_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=2e−04)
IZUMO4 | 215734_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3629)
IZUMO4 | 215734_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5648)
IZUMO4 | 215734_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IZUMO4 | 215734_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0476
p=0.0708
p=0.4398
p=0.5508
p=0.0024
p=0.8437
p=0.5404
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2